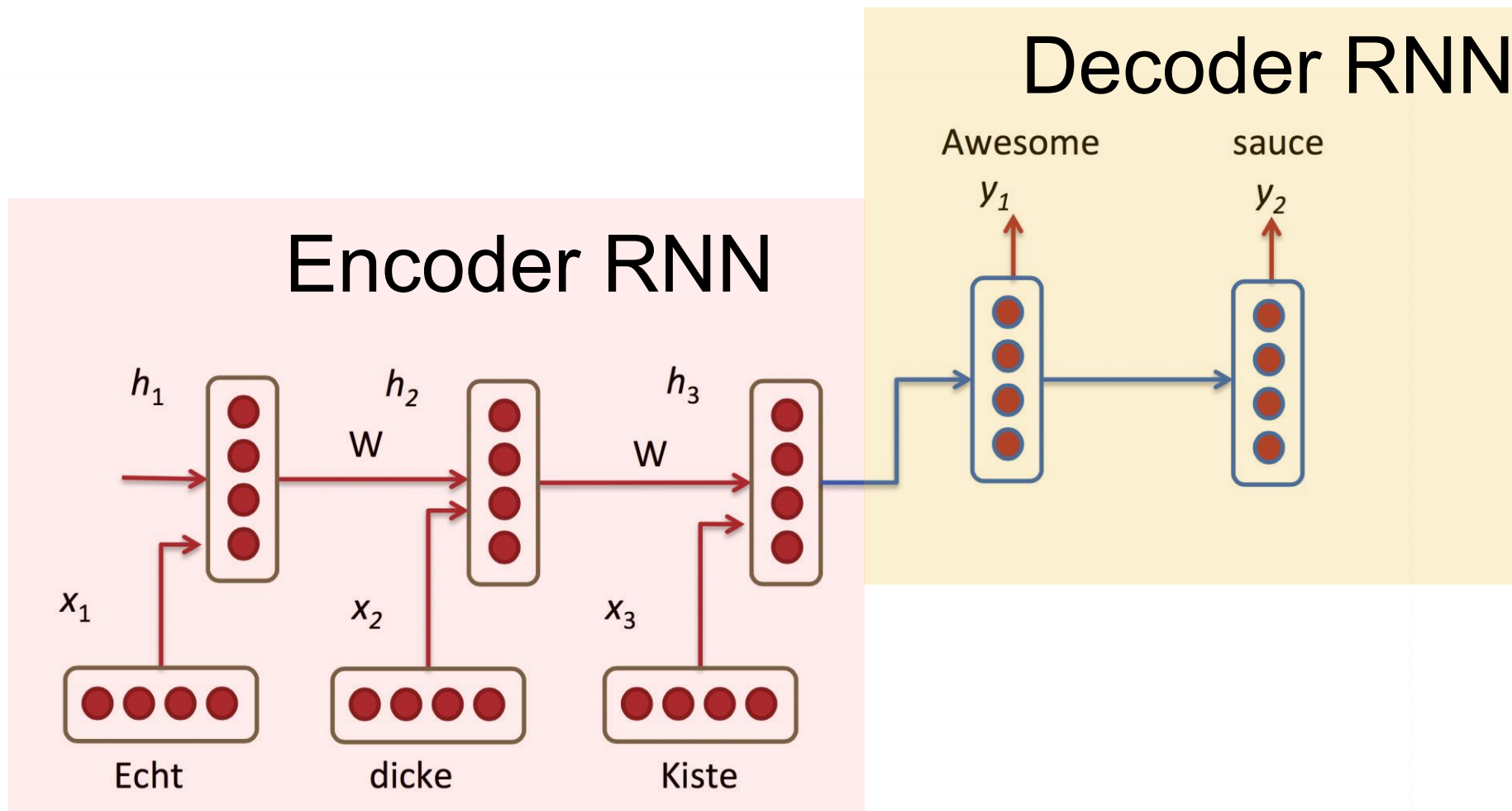


Sequence to Sequence

IT융합학부 | 진동섭

Seq2seq (Sutskever et al., 2014)



Seq2seq overview and applications

Encoder-decoder

Two RNNs (typically LSTMs or GRUs)

Can be deterministic or variational

Applications:

Machine translation

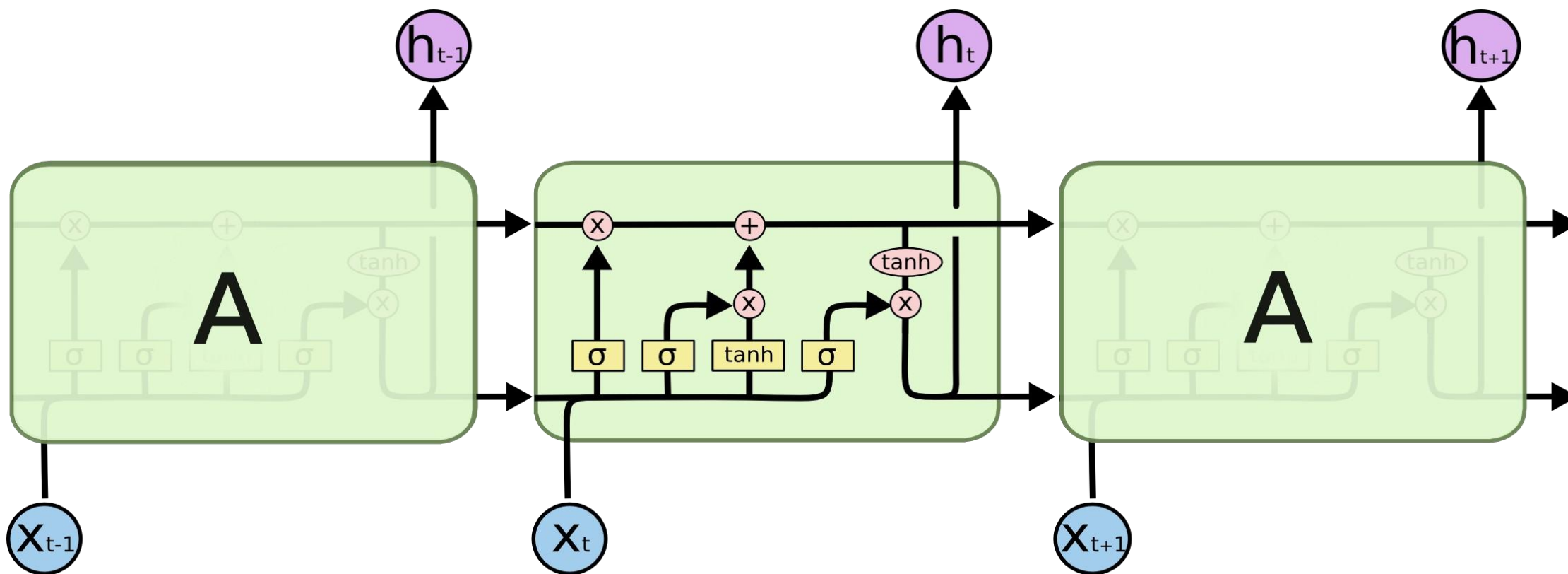
Question answering

Dialogue models (conversational agents)

Summarization

Etc.

LSTM cell



Seq2Seq

Source sequence $\mathbf{x} = (x_1, x_2, \dots, x_{|x|})$ represented as word embedding vectors

Target sequence $\mathbf{y} = (y_1, y_2, \dots, y_{|y|})$

At the end of the encoding process, we have the final hidden and cell states

Hidden state initialization:

Set the initial states of the decoder $\mathbf{h}_1^{(\text{tar})}$ and $\mathbf{c}_1^{(\text{tar})}$ to $\mathbf{h}_{|x|}^{(\text{src})}$ and $\mathbf{c}_{|x|}^{(\text{src})}$

Seq2seq (cont.)

At each step of the decoder, compute

$$\mathbf{h}_j^{(\text{tar})} = \text{LSTM}_{\theta}(\mathbf{h}_{j-1}^{(\text{tar})}, \mathbf{y}_{j-1})$$

$\mathbf{y}_{j-1}$ – ground truth previous word during training (“teacher forcing”), and previously predicted word at inference time.

θ – parameters (weights) of the network

Seq2seq (cont.)

Predicted word at time step j is given by a softmax layer:

$$p(y_j) = \text{softmax}(W_{\text{out}} \mathbf{h}_j^{(\text{tar})})$$

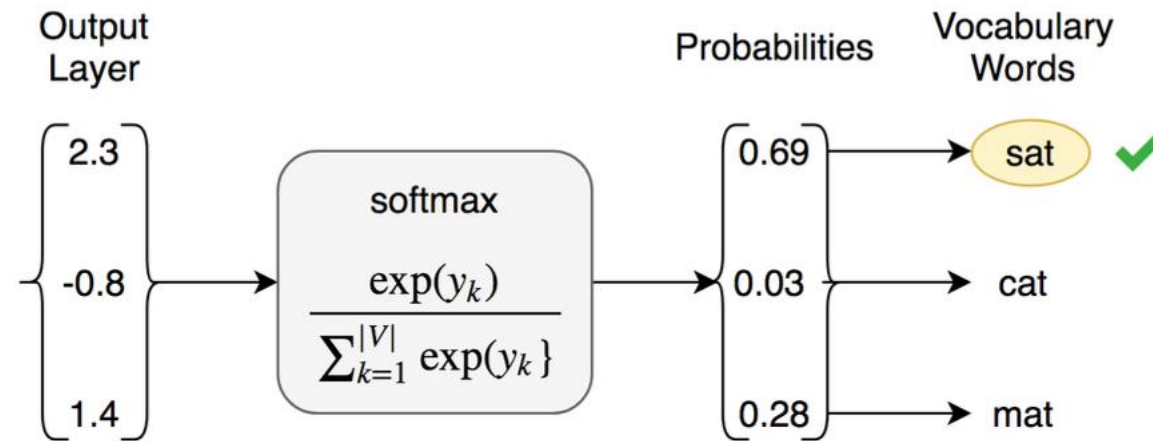
W_{out} is a weight matrix

Softmax function:

$$\text{softmax}(y_{jk}) = \frac{\exp y_{jk}}{\sum_{k=i}^{|V|} \exp y_{jk}}$$

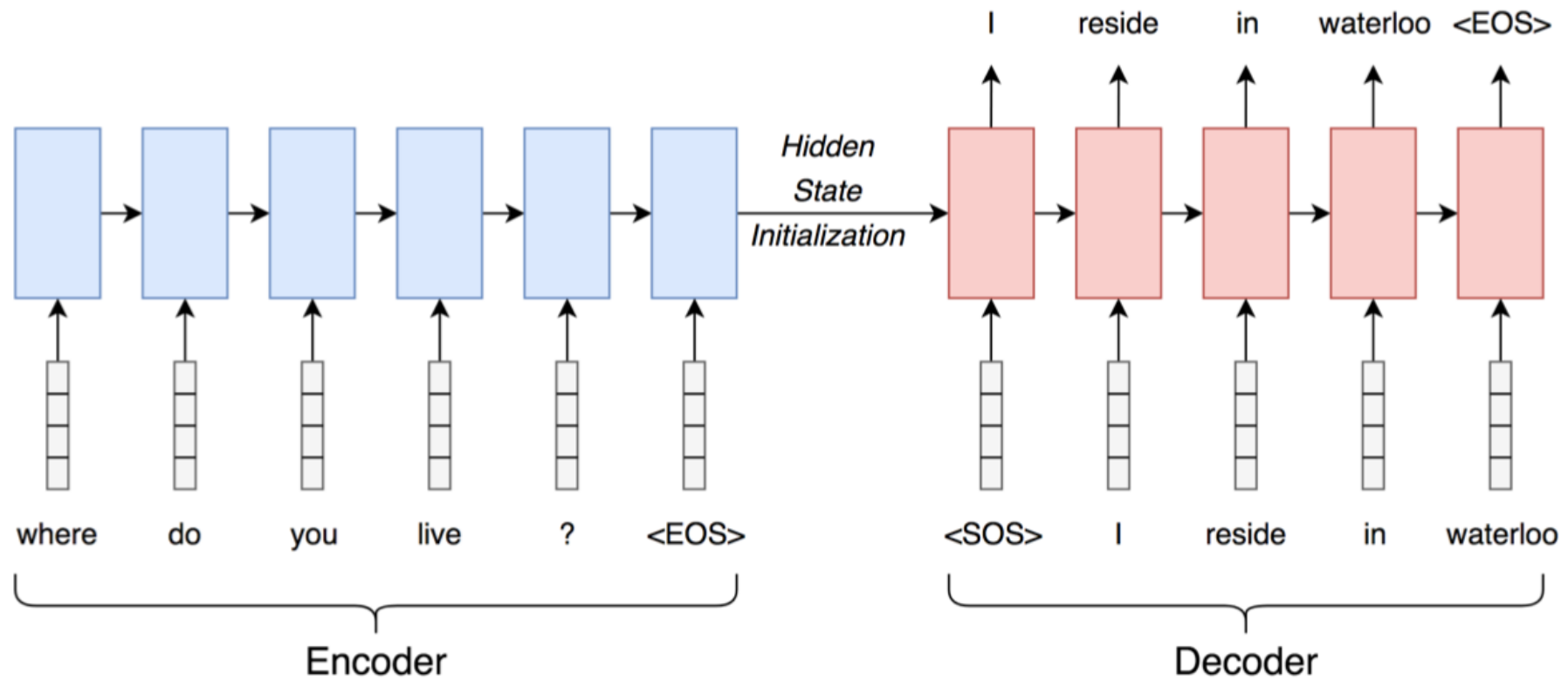
y_{jk} is the value of the k_{th} dimension of the output vector at time step j

Softmax example



Source: (Bahuleyan, 2018)

Seq2seq model



Selecting the word at each time step of the decoder

Greedy search: select word with the highest $p(y_i)$ given by the softmax layer

$$p(y_j) = \text{softmax}(W_{\text{out}} \mathbf{h}_j^{(\text{tar})})$$

Beam search: choose k words with the highest $p(y_i)$ at each time step.

k – beam width (typically 5-10)

Seq2seq model with attention

